

Week 8 Assignment: Linear Regression with Batch Gradient Descent

Raw NumPy, normalization, convergence, and numerical defenses

CMPSC 480 - Student Question Deck

Assessment overview

Build a vectorized regression pipeline and compare it with a stable closed-form oracle.

Coverage: Week 8: linear regression, MSE, gradient descent, and scaling

Points

10 total

Expected time

6-8 hours

Submission

Repository plus convergence report

Learning objectives

- ▶ Fit and reuse training-only normalization statistics.
- ▶ Implement a vectorized MSE gradient.
- ▶ Handle an intercept consistently.
- ▶ Stop on bounded, reproducible convergence criteria.
- ▶ Compare predictions with a pseudoinverse reference.

Requirements checklist

- ▶ Use Python 3.11 and NumPy; do not use a library that implements the assessed algorithm.
- ▶ Keep data, model, optimization, and evaluation responsibilities behind explicit interfaces.
- ▶ Validate shapes, ranges, finite values, empty inputs, and terminal or degenerate cases.
- ▶ Use deterministic seeds and record every configuration used for reported evidence.
- ▶ Include focused tests and a README with exact reproduction commands.

Task 1 - Data and normalization (2 points)

Validate X and y and fit z-score scaling on training features.

- a. Specify fit and predict shape rules and bias handling.
- b. Handle a zero-variance feature.

[1 points]

[1 points]

Task 2 - Vectorized optimizer (4 points)

Implement full-batch gradient descent with cost history and finite guards.

- a. Derive and implement the gradient of one-half mean squared error. [2 points]
- b. Compute one update for $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $y = [1, 3]$, $w = [0, 0]$, $\alpha = 0.5$. [1 points]
- c. Define divergence and stopping checks. [1 points]

Task 3 - Oracle comparison (2 points)

Compare learned predictions with NumPy pseudoinverse least squares.

- a. State why the pseudoinverse is preferable to explicitly inverting $X^T X$. [1 points]
- b. Define a comparison for a collinear dataset. [1 points]

Task 4 - Experiments and tests (2 points)

Evaluate learning rate and scaling while covering small and degenerate inputs.

- a. Compare two learning rates with and without scaling. [1 points]
- b. Test one feature, $m < n$, constant feature, empty input, and nonfinite data. [1 points]

Edge cases and defensive programming

- ▶ A constant feature.
- ▶ One training example.
- ▶ More features than examples.
- ▶ Learning rate causes nonfinite loss.
- ▶ Predict is called before fit.

Submission instructions

- ▶ Submit source, tests, README, configuration, and the requested report.
- ▶ Provide one command that reproduces the primary result from a clean environment.
- ▶ Exclude credentials, generated caches, and unlicensed data.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
Data and normalization	2	Stored means/scales and preprocessing tests.
Vectorized optimizer	4	Trainer, gradient tests, and convergence trace.
Oracle comparison	2	Tolerance checks on several synthetic datasets.
Experiments and tests	2	Controlled table and focused test suite.

Reflection prompt

In 150-250 words:

Explain one case where the loss curve looked plausible but an oracle or invariant exposed a defect.